

This is *almost* regular

Ian Mitchell
2025-11-21

1. Introduction

This is just the regular style, but with Arimo instead of Inter. I have a weird kind of neurosis instilled by Matthew Butterick about using “screen fonts” on something meant for a printed page, so I’m using the regularish style to get around that.

1.1. Some more stuff

1.1.1. Great news. This is like typing but with other stuff.

2. Standard demo

The following text is a collection of *fragments* taken from Lieb, Schultz, Mattis (2004/1961).¹ Please, go read the actual paper, it’s great.

2.1. Formulation

The first model consists of N spin $1/2$ ’s (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i [(1 + \gamma) S_i^x S_{i+1}^x + (1 - \gamma) S_i^y S_{i+1}^y],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy -plane. Because the Hamiltonian only involves the x - and y - components of the spin operators, we call this model the XY model.

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + i S_i^y \text{ and } a_i = S_i^x - i S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = (a_i^\dagger + a_i)/2; S_i^y = (a_i^\dagger - a_i)/2i; S_i^z = a_i^\dagger a_i - \frac{1}{2}$$

¹Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) Condensed Matter Physics and Exactly Soluble Models. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i \left[(a_i^\dagger a_{i+1} + \gamma a_i^\dagger a_{i+1}^\dagger) + \text{h.c.} \right].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k \Psi_0 = 0, \text{ all } k.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k.$$

In the limit $N \rightarrow \infty$, the sum can be replaced by an integral giving

$$\begin{aligned} \frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} [1 - (1 - \gamma^2) \sin^2 k]^{1/2} dk \\ &= -\frac{1}{\pi} \mathcal{E}(1 - \gamma^2), \end{aligned}$$

where $\mathcal{E}(k^2)$ is one of the complete elliptic integrals (13).